

mono = what you type in Excel · bold = the answer · red box = a trap · green box = the rule · course spelling NORMDIST / NORMINV / NORMSDIST / NORMSINV (.DIST/.INV forms are identical).

A · THE FIVE METHODS — when to use, keywords in the question, how, Excel

1. Expected value of scenarios (the mean)

When: outcomes with a chance each (a table, or “x% chance” in text); the end of every tree.

Keywords: expected value / profit, on average, with probability, chance of, fair price, scenarios

How: list outcomes; probabilities add to 1; multiply outcome × probability; add.

Ex: flats 0/1/2/3, P .1/.3/.3/.3 → $0(.1)+1(.3)+2(.3)+3(.3) = 1.8$.

Houses 145/150/155, .2/.6/.2 → **150**.

Excel: =SUMPRODUCT(A2:A5, B2:B5)

2. Standard deviation of scenarios (the risk)

When: same table, asked for risk / variability / how much it swings. **Mean FIRST.**

Keywords: standard deviation, variance, risk, volatility, riskier, how much it varies

How: (outcome – mean)² × probability, add = variance; square root = stdev.

Ex: $.1(0-1.8)^2+.3(1-1.8)^2+.3(2-1.8)^2+.3(3-1.8)^2 = .324+.192+.012+.432 = .96 \rightarrow \sqrt{.96} = .9797$. Forgot the root? Huge number, squared units.

Excel: =SQRT(SUMPRODUCT((A2:A5-mean)^2, B2:B5))

Y = a + bX: E = a + b·E(X); sd = |b|·sd(X). $-50+200X \rightarrow 310, 195.96$. a never changes the spread. Tiered payoff ≠ a+bX: rebuild the table on the payoff (bonus 0/15/52.5/82.5 → 45, 30.19).

3. Normal: z and probability

When: “normally distributed” + a mean and stdev. **Forward:** value → %. **Backward:** % → value.

Keywords: less than, more than, between, at least, percentile, 90% chance, how many to produce, VaR, top 10%, what score

How: forward: $z = (\text{value} - \text{mean}) / \text{sd} \rightarrow \text{NORMSDIST}(z) = \% \text{ below}; \text{above} = 1 \text{ minus}; \text{between} = \text{larger minus smaller}$. Backward: % BELOW → NORMSINV → z → value = mean + z·sd.

Ex: N(80,10): P(X<90) z=1 → **.8413**; P(>90) **.1587**; P(75–90) .8413–.3085 = **.5328**; top 10% NORMINV(.90,80,10) = **92.8**. VaR N(7,10): NORMINV(.05,7,10) = **–9.45%**. Demand N(1200,130): P(<1350) .8749; 95% stock 1413 → 1414.

Excel: =NORMSDIST(z) =NORMSINV(p)

=NORMDIST(x, μ, σ, TRUE) =NORMINV(p, μ, σ)

4. SAXBY: two variables D = a·X + b·Y

When: TWO variables: two demands, two stocks, a weighted score, pooling, or X compared with Y. Normal in → normal out, then method 3.

Keywords: portfolio, invest in each, total / combined, correlated, independent, pooling, higher than, exceeds, more than twice, weighted

How: 1) a, b from the question (units, dollars, weights; X>Y → a=1, b=–1; “twice” → b=–2). 2) mean = aμX + bμY (no p). 3) sd = $\sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y}$. 4) method 3.

Ex: walk-in N(100,10), phone N(150,15), p .8. Total: 250; $\sqrt{(100+225+240)} = 23.77$; stock 99%: NORMINV(.99,250,23.77) = 305.3 → **306**. X–Y: –50, $\sqrt{(100+225-240)} = 9.22$; P(X>Y) = 1–NORMDIST(0,–50,9.22,1) ≈ **0**.

Excel: =a*mX+b*mY

=SQRT(a^2*sX^2+b^2*sY^2+2*a*b*rho*sX*sY)

5. Trees, “at least one”, Bayes

When: stages, several “ifs”, repeated years, “given the test was positive”.

Keywords: if positive / negative, survey then drill, over N years, at least one, never, given that, false alarm, default

How: along a path **multiply** (AND); across paths **add** (OR); endpoints sum to 1. At least one = $1 - (1-p)^N$. Bayes: $P(A|+) = \text{path}(A \text{ and } +) \div \text{all “+” paths}$.

Ex: AAA .0002 × 25 yrs: $1 - .9998^{25} = .004988$. HIV: $.006 \cdot .999 \div (.005994 + .00994) = .376$. \$1 × 1.1/× 0.9 twice: E **1.04**, sd **.141**.

Excel from scratch

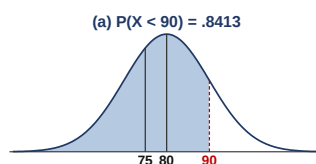
SUMPRODUCT (A2 : A5, B2 : B5)	row×row then add
SQRT (x) x^2 \$B\$7	root · square · \$ locks when dragging
NORMSDIST (z) / NORMSINV (p)	z → % below / % below → z
NORMDIST (x, m, s, TRUE) / NORMINV (p, m, s)	value → % below / % → value. Always TRUE
1-NORMDIST (. . .)	% ABOVE
AVERAGE MEDIAN STDEV CORREL SKEW KURT	raw data (KURT is excess: +3)
BINOMDIST (k, N, p, FALSE)	exactly k of N; TRUE = at most k

z (area left): 80% .8416 · 90% 1.2816 · 95% 1.6449 · 97.5% 1.96 · 99% 2.3263. Above: 1 minus. Below mean: negative. $\pm 1\sigma$ 68.3% $\pm 2\sigma$ 95.4% $\pm 3\sigma$ 99.7%. Raw data: handout says STDEV.S (n–1); **practice solution uses STDEV.P — check which the marker wants.**

B · NORMAL — draw the bell, shade, then Excel

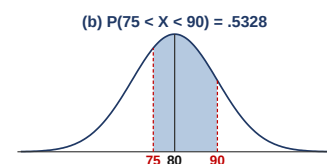
Worked: grades $\mu=80, \sigma=10$ — draw, shade, then one Excel line

Given the VALUE, want a % → z → **NORMSDIST**. Given the %, want the value → **NORMSINV** → mean + z·sd. Both use the area to the **LEFT**. Shaded on the right? **1 minus**.



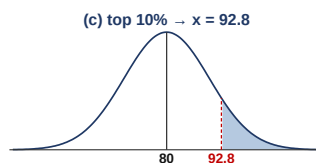
$z = (90-80)/10 = 1 \rightarrow \text{NORMSDIST}(1)$. Above the mean ⇒ 50% ✓

=NORMDIST(90, 80, 10, TRUE)



larger minus smaller: .8413 – .3085

=NORMDIST(90, 80, 10, TRUE)
–NORMDIST(75, 80, 10, TRUE)



top 10% above = 90% below: NORMSINV(.9) = 1.2816; $x = 80 + 1.2816 \times 10$

=NORMINV(.9, 80, 10)

Mistakes: forgetting TRUE · NORMINV(0.10) instead of 0.90 for “top 10%” · no 1 minus for “more than” · % and decimals mixed · shading the wrong side. **Estimate first:** is x above or below the mean? Then is the answer above or below 50%? If the picture disagrees with the number, the number is wrong.

Check two ways: (a) z + NORMSDIST, and NORMDIST directly. (c) NORMSINV + mean+z·sd, and NORMINV directly.

Variations (one Normal variable)

If the exam says	Type	μ80 σ10
more than / at least 90	=1 - NORMDIST(90, μ, σ, TRUE)	.1587
value with 5% below (VaR, worst)	=NORMINV(0.05, μ, σ)	63.55
top 10% / 90th percentile	=NORMINV(0.90, μ, σ)	92.82
stock for a 95% chance	=NORMINV(0.95, m, s) → round UP	N(1200,130): 1414
within k stdevs	=NORMSDIST(k) - NORMSDIST(-k)	k=2: .9545
which of two was rarer	z each, compare z	Shanghai 1.54 < NYSE 2.21
what percentile am I	=NORMSDIST(z)	z 1.5 → 93rd
chance of a loss	=NORMDIST(0, m, s, TRUE)	N(8.79,15.75): .288

C · SCENARIOS, TREES, BAYES

Scenario table → E(X), V(X), sd (flats 0/1/2/3, P .1/.3/.3/.3)

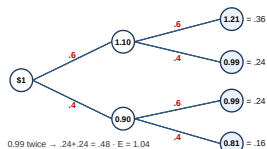
- Check $\Sigma P = 1$.
- Mean: $\Sigma x \cdot P = 0(.1) + 1(.3) + 2(.3) + 3(.3) = \mathbf{1.8}$.
- Each $(x - 1.8)^2$: 3.24, .64, .04, 1.44.
- Variance: $3.24(.1) + .64(.3) + .04(.3) + 1.44(.3) = \mathbf{.96}$.
- sd = $\sqrt{.96} = \mathbf{.9797}$.

Profit -50+200X: E 310, sd 195.96 — or put the profits (-50, 150, 350, 550) in the outcome column with the same P. Houses X 145/150/155 (.2/.6/.2) vs Y 147/150/153 (.4/.2/.4): both E 150; sd 3.16 vs 2.68 → Y is the safer buy: compare E first, then sd. Risk is sd, not the chance of a loss alone. Chuck-a-luck -1/1/2/10 with .579/.347/.069/.005: E $\mathbf{-\$0.46}$, sd 1.288 — every casino game has $E < 0$.

=SUMPRODUCT(A2:A5, B2:B5) mean

=SQRT(SUMPRODUCT((A2:A5 - mean)^2, B2:B5)) sd

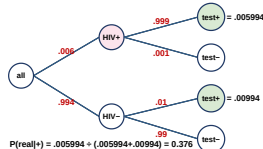
Tree · at least one · Bayes



Probability ON the branch; multiply along (.6×.6 = .36); same endpoint twice → add (.24+.24 = .48); endpoints sum to 1; then methods 1 and 2 on the endpoints: E **1.04**, sd **.141**.

P(at least one in N) = $1 - (1 - p)^N$

ONE never-path: compute it, subtract. AAA: $=1 - (1 - 0.0002)^{25} = \mathbf{.004988}$. Never add up the "at least one" cases.



Split by **condition** first (.006/.994), then by **test**. Circle the paths that test +: .006×.999 = .005994 and .994×.01 = .00994. Answer = wanted path ÷ all + paths = $.005994 / (.005994 + .00994) = \mathbf{.376}$. A 99.9%-accurate test on a .6% condition is mostly false alarms: the healthy branch is huge.

$P(A|+) = P(A) \cdot P(+|A) \div [P(A)P(+|A) + P(\text{not } A)P(+|\text{not } A)]$

Variations (scenarios and trees)

If the exam says	Type	Answer
5 scenarios	extend ranges A2:A6 etc.	
profits in text (40 w/ 25%, -10 w/ 75%)	type as a table; $\Sigma P = 1$	E 2.5, sd 21.65
min. chance to break even (E = 0)	p·gain + (1-p)·loss = 0	40p -10(1-p) → 20%
compare two decisions	larger E; tie → smaller sd	

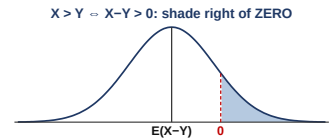
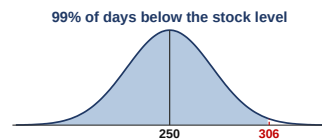
a survey / test costs money first	tree; subtract cost on EVERY path	
P(one path) / P(A and B) indep.	multiply branches / $P(A) \times P(B)$	.5×.5 = 25%
P(A or B)	$P(A) + P(B) - P(A \text{ and } B)$	
exactly k in N	=BINOMDIST(k, N, p, FALSE)	

D · SAXBY — two variables, then one bell

Total demand (a=1, b=1) and P(X > Y) (a=1, b=-1) — walk-in N(100,10), phone N(150,15), ρ .8

- Right variable: $D = X + Y \rightarrow a = 1, b = 1$.
- Mean: $100 + 150 = \mathbf{250}$ (p does not enter).
- Variance: $1^2 \cdot 100 + 1^2 \cdot 225 = 325$; cross term $2 \cdot 1 \cdot .8 \cdot 10 \cdot 15 = \mathbf{+240}$; sd = $\sqrt{565} = \mathbf{23.77}$.
- Stock for 99% of days: NORMINV(.99, 250, 23.77) = 305.3 → **306**. Combine FIRST, percentile AFTER — never add two separate 99% levels.

=SQRT(a^2*sX^2+b^2*sY^2+2*a*b*rho*sX*sY)
=NORMINV(0.99, 250, 23.77)



- Move everything to one side: $X - Y > 0 \rightarrow a = 1, b = -1$ ("more than twice" → b = -2).
- Mean: $100 - 150 = \mathbf{-50}$.
- Variance: $100 + 225 - 240 = 85 \rightarrow$ sd **9.22**. b² kills the sign; the cross term flips.
- $P(D > 0) = 1 - \text{NORMDIST}(0, -50, 9.22, \text{TRUE}) \approx \mathbf{0}$: zero sits 5.4σ above the mean.

=1 - NORMDIST(0, -50, 9.22, TRUE)

Mark **ZERO** on the D curve, not the mean. Independent (ρ=0)? The cross term simply disappears.

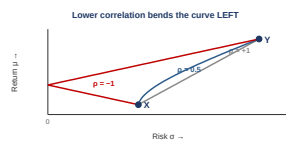
Variations (SAXBY)

If the exam says	Type	Demand numbers
total exceeds 280	=1 - NORMDIST(280, μD, σD, TRUE)	10.3%
95% service level	=NORMINV(0.95, μD, σD)	289.1 → 290
independent	ρ = 0	sd $\sqrt{325} = 18.03$
2 of X and 3 of Y per order	a = 2, b = 3	mean 650
X more than twice Y	a = 1, b = -2, then P(D>0)	
dollars invested (returns given)	a, b = \$ amounts; result in \$	\$10K/\$10K, 50%/30%, p -.4: \$8,000, \$4,183
weights as % of portfolio	a=.5, b=.5; result in %	40%, 20.9%
weighted score .35 / .65, both N(100,16)	a=.35, b=.65; NORMDIST(121.6,100,σD,TRUE)	ρ1: .9115 · ρ.5: sd 14.1, .937 — lower ρ = higher pct
covariance given	ρ = cov/(σXσY)	
pooling: shared sd smaller?	σD vs σX + σY	23.77 < 25

E · PORTFOLIO $P = wX + (1-w)Y$ — three targets, the frontier, pooling

$$E(P) = w\mu_X + (1-w)\mu_Y$$

$$\sigma(P) = \sqrt{(w^2\sigma_X^2 + (1-w)^2\sigma_Y^2 + 2w(1-w)\rho\sigma_X\sigma_Y)}$$



$\rho +1$: straight line, no benefit · 0: bends left · -1 : reaches zero risk. Leftmost point = min-variance portfolio.

① **Target return** (linear): $w = (\text{target} - \mu_Y) / (\mu_X - \mu_Y)$. 10%/20%, target 15% → .5; 15%/25%, 16% → .9.

② **Target risk** (quadratic, 2 roots): $A = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$, $B = 2\rho\sigma_X\sigma_Y - 2\sigma_Y^2$, $C = \sigma_Y^2 - \text{target}^2$, $w = (-B \pm \sqrt{(B^2 - 4AC)}) / 2A$, keep the root in $[0,1]$. 15%/35%, $\rho .5$, 20% → .60.

③ **Min risk**: $w^* = (\sigma_Y^2 - \rho\sigma_X\sigma_Y) / (\sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y)$. Houses 3.16/2.68, $\rho .5$ → $w^* .44$, σ 1.45 < both. Outside $[0,1]$ = short selling → boundary.

Beat the better return? NO. Below the worse? NO. Lower sd than both? YES (portfolio effect). **Pooling**: GPUs $\mu 20$ $\sigma 10$ each, $\rho .2$, 3 σ cover: separate 50+50 = 100; pooled 40 + 3 $\times\sqrt{240}$ = 86.5. Means add; sd shrinks; at $\rho=1$ no saving.

F · THEORY, LECTURES 1–5

L1 · Dispersion and z

- **Stdev** = typical distance from the mean; same units as data. Mean alone is not enough (68–72 vs 40–100 both mean 70; sd 1.41 vs 21.21).
- **Variance** = mean squared deviation; we square because deviations cancel to zero; root back to units. 2,5,7,9,10: $\Sigma(x-\bar{x})^2$ 41.2, S^2 10.3, S 3.21.
- **Mean** dragged by outliers (Bill Gates in the bar); **median** resistant. Replace 10 by 10000: median 7, mean 2004.6.
- $z = (\text{value} - \text{mean}) / \text{sd}$: unitless, compares different things. $|z| < 1$ normal, 2 unusual, > 3 very rare. Works for ANY shape; only $z \rightarrow \%$ needs Normality.

L2 · Shape and correlation

- **Right skew**: tail RIGHT, **mean > median** (income 106.5k vs 83.7k; box office; market caps). **Left**: tail LEFT, **mean < median** (easy grades, satisfaction). Name follows the TAIL. $|\text{skew}| \geq .5$ significant.
- **Kurtosis** = sharper peak AND heavier tails (more extremes both ways). Normal = 3; **Excel KURT is excess (Normal 0): add 3**. Heavy tails → Normal understates disasters (S&P March 2020 = -6σ). Kurt 5 vs 3: Normal approx riskier for the 5.
- ρ in $[-1,1]$; 0 = no LINEAR relation (a parabola can be 0). Sign = direction, size = tightness. **Not the slope** (.30 $\neq$ 30¢ per \$), **not causation** (feet vs education .849). Breaks in crises (LTCM). $\text{cov} = \rho\sigma_X\sigma_Y$.

L3 · Probability and $E[X]$

- RV = outcomes + probabilities summing to 1. **$E[X]$** = weighted average = long-run average = fair value = balance point; need not be a possible value (die: 3.5). Compare E first, then risk.
- Complement: $P(\text{at least one}) = 1 - P(\text{none})$. Tree: multiply along, add across; endpoints net of what that path cost. All casino games $E < 0$.
- **Bayes**: split by condition FIRST, then test; wanted path ÷ all paths with that result. 99.9% test on a .6% condition → $P(\text{real} | +) = .38$: the healthy branch is huge.

L4 · SAXBY and diversification

- a, b = context multipliers (weights .4/.6, units 1/2, scores .35/.65, pooling 1/1, difference 1/–1); need not sum to 1. **ρ changes the risk, never the mean.**
- $\rho=+1$: sd is the plain sum, no diversification. $\rho=0$: cross term gone. $\rho=-1$: minimum, can reach zero. **Stdevs do not add**: variance first, root after; weights enter SQUARED; decimals or plain numbers, never mixed.
- **Portfolio effect**: a mix can be safer than EITHER asset (houses 40/60, $\rho .5$: 1.47; best .44: 1.45). Efficient frontier (Markowitz): lower ρ bends it LEFT; leftmost = min variance.

L5 · Normal

- Bell fixed by μ, σ ; **probability = area**; $P(\text{exact}) = 0$, so $<$ and $\leq$ agree. $Z \sim N(0,1)$; NORMDIST/NORMINV use the area LEFT. 4 types: below, above (1–), between (big–small), inverse.
- VaR $N(7,10)$: -9.45% = “with 95% confidence the yearly loss will not exceed 9.45%”. S&P $N(8.79,15.75)$: $P(<0)$.288, $P(0-12)$.292, worst 5% –17.11.
- **Normal + SAXBY**: $aX+bY$ of normals is normal; combine, then ONE bell. G&T .35/.65, $N(100,16)$, Ann 121.6: $p1$ sd 16 → .9115; $p.5$ sd 14.1 → .937. Lower ρ = higher percentile. Sales data hide unmet demand (censored).

Question → tool

If you see	Use
raw numbers; “describe”, skew	AVERAGE / MEDIAN / STDEV / SKEW / KURT+3
“more unusual”, percentile	z each, $ z $; NORMSDIST
two columns, true/false about p	CORREL; sign, not slope, not cause
scenario table, “expected”, “std dev”	methods 1, 2
“profit = fixed + per unit”	$a + bE(X)$, $ b sd$
“if positive”, survey, over N years	tree; $1 - (1-p)^N$
“given the test was +”	Bayes
“normally distributed”, less/more/between	value → z → %
“find x”, “90% chance”, how many, VaR	% → NORMINV
two variables, weights, portfolio, correlated	SAXBY → Normal
“A more than twice B”, “X higher than Y”	$P(X-bY > 0)$
target return / risk / safest mix	①②③
pooling, k stdevs of cover	SAXBY $a=b=1$; $\mu+k\sigma$

Checklist

1. One sentence: **below, above, between, or find the value?**
2. Scenarios: $\Sigma P = 1$, each profit on its own row.
3. Two variables: a, b from the question; mean, then SAXBY sd, then one bell.
4. “More than / exceed”: **1 minus**. NORMINV: area BELOW (“top 10%” → .90).
5. Estimate first (above/below the mean? $>$ or $<$ 50%?); check two ways.
6. Same scale for μ, σ ; all decimals; units round UP; took the root?
7. Show formula, numbers, result with units in Canvas.